



Applications of natural and synthetic fiber reinforced polymer in infrastructure: A suitability assessment

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ABSTRACT

Using fiber-reinforced polymer (FRP) composites in engineering infrastructures has effectively resolved significant problems related to restoring existing and deteriorating infrastructures. The usage of FRP composites such as synthetic FRP, has received much attention in recent years. However, there has been a surge in demand for natural FRP due to its mechanical and durability performance, eco-friendliness, lightweight, life-cycle superiority, biodegradability, and low cost. However, the applications of natural FRP in infrastructures are limited due to the lack of knowledge of their performance. Thus, this study conducted a comprehensive review of the strength, durability, fire, sustainability and trending development of FRP applications with a focus on civil infrastructure. The suitability assessment review addresses the mechanical, fire, durability and sustainability performance of synthetic and natural FRP and their composites. Then, the review discusses recent development in this field. In particular, the revolutionary creation of hybrid FRP and their performance and application infrastructure were discussed together with constraints and future research directions. The review highlights that the hybrid FRP is the sustainable option to use civil infrastructure applications to improve structural, fire, durability and sustainability performance.

Abbreviations

FRP	Fiber-reinforced polymer
SFRP	Synthetic fiber-reinforced polymer
NFRP	Natural fiber-reinforced polymer
HFRP	Hybrid fiber-reinforced polymer
EBR	Externally bonded reinforcement
NSM	Near-surface mounted reinforcement
CFRP	Carbon fiber-reinforced polymer
GFRP	Glass fiber-reinforced polymer

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AFRP	Aramid fiber-reinforced polymer
BFRP	Basalt fiber-reinforced polymer
WoS	Web of Science
T_g	Transition temperature of glass
HSCG	High strength cement grout
ECC	Engineered cementitious composites
PVA	Polyvinyl alcohol
PET	Polyethylene terephthalate
TRM	Textile reinforced mortar
PLA	Polylactic acid

1. Introduction

Fiber Reinforced Polymer (FRP) composites have gained attention in terms of restoring existing and deteriorating infrastructures in the last two decades [1]. FRP composites enhance structures' durability while having lower self-weight and facilitating easy installation. Shear and flexural strengthening of the infrastructures using FRP was found as a successive solution through many research [2,3]. The strengthening is generally employed to carry additional loads, enhancing the capacity loss due to deterioration, imperfect initial design, new standards, accidents such as fire or seismic events, aging of materials, and construction deficiencies due to lack of maintenance [4–6]. Two primary techniques such as externally bonded reinforcement (EBR) and near-surface mounted reinforcement (NSM) [7] are used to strengthen infrastructure with FRP. The EBR technique is a way of applying FRP as full wrapping, bonding on two lateral surfaces, inclined wrapping or U-wrap using epoxy adhesive [8,9]. In the NSM FRP strengthening method, grooves are carved in the cover of reinforced concrete (RC) members at the beginning. Then the FRP bars, laminates or strips are subsequently implanted with an epoxy adhesive into the grooves, as shown in Fig. 1 [10,11]. EBR-FRP reinforcement is a typical method of repair and strengthening prevailing over the last two decades [12]. NSM-FRP reinforcement has recently received much attention due to its advantages over EBR, including a larger bond surface, higher tensile strength, and higher resistance to peeling-off [13].

The fibers generally used in the construction of infrastructure can be primarily classified into two types, which are synthetic fibers and natural fibers. The synthetic fibers being addressed throughout this paper are non-plant-based fibers such as carbon, glass, basalt, and aramid fibers [14]. These fibers have led to the creation of several Synthetic Fiber Reinforced Polymer (SFRP) materials such as carbon FRP (CFRP), glass FRP (GFRP), aramid FRP (AFRP), and basalt FRP (BFRP) through a pultrusion technique [15,16]. The pultrusion technique is widely used as composite synthesis method in which the fibers are pulled through a heated die while pressure is being applied to melt the resin and impregnate it into the fiber reinforcement [17]. SFRP materials are extremely flexible and simple to install, making them ideal for flexural, shear, torsional, and axial retrofitting applications. The SFRP composites are applied in the form of laminates, rods, dry fibers, or sheets [6].

SFRP composites also have grown in popularity due to their non-corrosive properties, as well as their high longitudinal tensile strength, stiffness, strength-to-weight ratio, insect and fungus resistance, and chemical attack resistance [18–20]. However, the primary issue with SFRPs is the non-environmentally friendly manufacturing of synthetic fibers and the subsequent decomposition issues. Glass and carbon fiber processing create significant health risks and can even result in human death [19–21]. Moreover, the loss of mechanical characteristics of composites and bond strength at elevated temperatures is a significant issue of FRP strengthening [22,23]. This necessitated the use of environmentally friendly fibers, which led to the introduction of natural fibers.

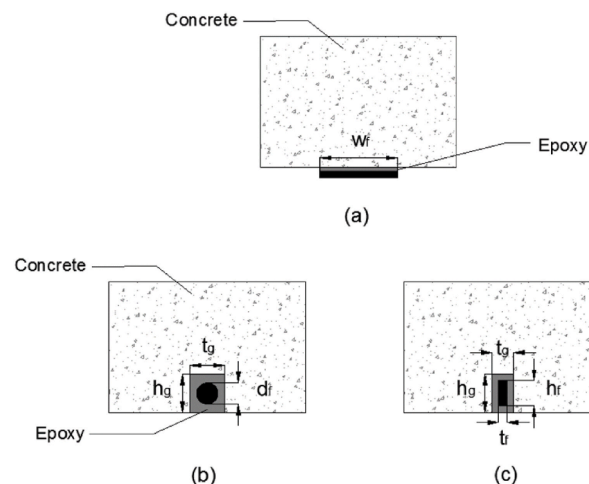


Fig. 1. Strengthening using EBR and NSM methods: a) EBR; b) NSM using a bar; and c) NSM using a laminate strip [11].

Natural fibers that are being discussed throughout this paper are plant-based and have numerous advantages over synthetic fibers, including low relative density, low cost, high impact resistance, and high flexibility, low specific gravity, less abrasiveness to equipment, fewer health hazards, process-friendly, lower greenhouse emissions, recyclability, and CO₂ neutrality [24]. Various types of natural fibers are used to develop the natural fiber reinforced polymer (NFRP) composites, which are banana, jute, hemp, flax, sisal, ramie and kenaf fibers [25]. Moreover, sisal and jute fibers have largely replaced glass and carbon fibers due to their ease of availability, low cost, and environmentally friendly nature [26]. A study by Ashik et al. [27] stated that NFRPs could be utilised instead of GFRPs in numerous infrastructure applications due to their eco-friendliness and lower density. Moreover, NFRP composites may also provide comparable mechanical property levels to SFRP composites [28]. NFRPs are used in many infrastructures as building materials in the form of straw bales, especially in the construction of buildings in the United States, Germany, etc. [29]. Despite the numerous benefits, natural fibers have significant drawbacks, including inhomogeneous quality and supply cycles, low water resistance and dimensional stability, rotting susceptibility, low impact strength, and excessive moisture absorption [30,31]. These limitations can be overcome by employing the hybridization process, which resulted in the development of Hybrid Fiber Reinforced Polymer (HFRP) composites.

HFRP composites are an emerging area in civil engineering that is growing rapidly for use in a various infrastructure applications [32]. HFRP composites have been shown to provide a balanced effect within the fibers used in the material [33]. The hybridization technique can improve the mechanical properties and reduce the constraints of single NFRP composites. The HFRPs are available in a combination of synthetic and natural fiber or natural and natural fiber [34,35]. The mechanical characteristics of HFRP composites are found to be improved with the addition of either synthetic fiber or natural fiber [30]. Further, the tensile, flexural, and impact properties of HFRP composites improved with the addition of synthetic fibers to natural fibers. Also, the moisture resistance and fire retardancy characteristics of natural fibers can be improved by combining natural fibers with synthetic fibers [36]. As a result, several HFRPs have been developed, including carbon/flax, ramie/glass, basalt/glass, kenaf/glass, flax/basalt, flax/Kevlar, jute/hemp, sisal/hemp, sisal/jute FRP [35]. HFRP is utilised in the building and construction industry, specifically partition and false ceiling panels, partition boards, wall, floor, window and door frames and roof tiles [36].

However, the application of SFRP, NFRP and HFRP in the infrastructure is limited due to the production cost, manufacturers, significant variability in the mechanical properties and limited knowledge of the durability, structural and fire performance. Therefore, this paper aims to review of the applications and recent advancement of SFRP, NFRP, and HFRP in infrastructure and their performance in terms of tensile, bond, fire, and durability. Fig. 2 illustrates the review contents and the sections where they were discussed in the paper.

2. Methodology

A detailed literature review of applications of FRP in infrastructure was investigated through journal articles. Two main databases are recognized to complete a bibliometric analysis, Web of Science (WoS) and Scopus. In this data collection used, several search

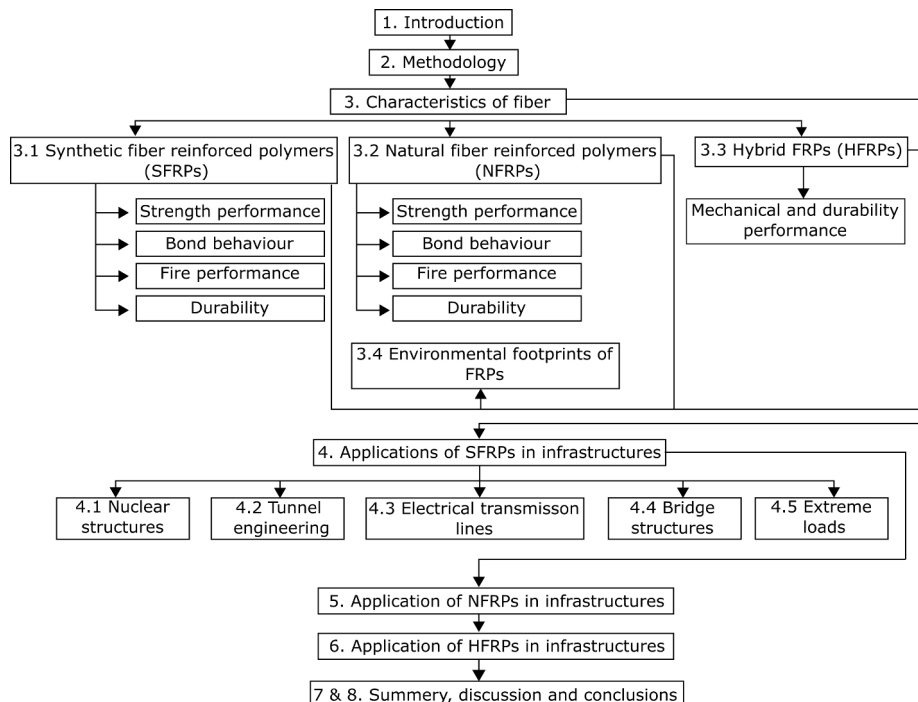


Fig. 2. The contents of the manuscript.

strings were listed in Table 1. To identify the recent development, the searches were limited to the journal published from the year 2012–2022. The investigation found 833 and 986 journal articles, respectively from Scopus and WoS. VOSviewer software was used to visualize the trend of the literature related to the fiber-reinforced polymer. Fig. 3 shows the keyword co-occurrence visualization diagram obtained using thresholds of 105 and 325 keywords, respectively Scopus and WoS in VOSviewer. The largest area of research conducted between the keyword represents by bigger circles in Fig. 3. The VOSviewer analysis revealed that the topmost regularly appearing keywords are “Fibre reinforced polymer” and “Reinforced concrete” and “Behaviour” and “Performance”. Furthermore, the scientific mapping shows that the aforementioned keywords have a larger circle than the other keywords, indicating that they are the most valuable terms in the current research topic. Fig. 3b indicates that other areas have not been the main focus investigated in infrastructure application. Therefore, this study intends to provide a comprehensive review of the applications of FRP in infrastructure.

3. Characteristics of FRPs

The FRP can be synthetic fiber reinforced or natural fiber reinforced or hybrid fiber reinforced polymer. Fig. 4 shows the most common synthetic and natural fibers that have been used to produce polymer composites.

3.1. Synthetic fiber reinforced polymers (SFRPs)

The SFRP have been widely used to strengthen concrete [37–43], steel [40,44] and timber [45–47] structures. Synthetic fibers can be categorised into three types such as carbon fibers, glass fibers (E-glass, S-glass and Z-glass) and aramid (aromatic polyamides, Kevlar 49) [14]. The polymer matrix (resins) and high-performance fibers (carbon, glass and aramid) are the two main components of SFRP composites [48]. The combination of polymer matrix with reinforcing fibers has resulted in the creation of several traditional SFRP materials, including carbon FRP (CFRP), glass FRP (GFRP), aramid FRP (AFRP), and basalt FRP (BFRP) [48,49]. Reinforcing bars made from materials are given in Fig. 5 [50], and those fibers are used as small discrete fibers as well depending on the application (Fig. 6). The most widely utilised SFRP products are prefabricated plates, bars, laminates sheets, and anchorages [14].

Textile fibers are another potential synthetic fiber for structural reinforcement in infrastructure applications [51]. Textile-reinforced composites are made up of a textile reinforcing phase and a matrix phase that is generally made up of a polymer. Textile materials include 2D or 3D woven fabrics, knitted fabrics, stitched fabrics, braids, nonwovens, and multiaxial fabrics used to produce textile fibers [51]. Textile reinforced polymers offer many advantages for infrastructures such as high toughness, damage tolerance, structural integrity, ease of handling, low fabrication cost, handle ability of the reinforcing materials, and appropriateness for net-shape production [52–55]. Textile-reinforced polymers can be used in place of steel in reinforced concrete and automotive applications in infrastructure [55].

3.1.1. Strength performance of SFRPs

The currently evolving FRP strengthening technologies have been significantly influenced by the high tensile strength of SFRP on reinforced concrete structures in most structural improvements in infrastructure. The tensile behavior is considered a fundamental property of FRP in its use in the concrete structure design. Further, it also contributes to the optimisation process of SFRP with respect to its application as reinforcement to concrete structural elements [56]. The SFRPs are used as internal reinforcing and exterior strengthening for RC structures. Due to the higher tensile capacity of the fibers used in polymer matrices, the SFRP concrete structures are considered to have superior tensile properties [56,57].

The SFRP composites consist of different orientations of fibers within the matrix, improving the tensile strength of SFRP composites. In the matrix where fibers are arranged in a straight, parallel and continuous manner, the maximum tensile capacity is found to be obtained parallel to the fiber direction [57]. Further, the tensile strength of composites with synthetic fibers in the parallel direction was about 20–40% higher than those of composites with fibers in the perpendicular direction. Moreover, the type of fiber, nature of the matrix, section interface, moisture absorption, fibers treatment, fillers or any additives and fabrication methods also affect the tensile strength of SFRP composites [58,59].

The range of tensile strength and elastic modulus of SFRPs are detailed in Table 2. This table shows that CFRP has a higher tensile strength than other FRPs. The tensile strength of SFRP composites is varied with exposed temperatures and strain rates [60–62]. Hawileh et al. [63] investigated the effect of temperature (i.e. 25–300 °C) on the tensile properties of CFRP and GFRP laminates. Their study found a reduction in the tensile strength of CFRP and GFRP at 250 °C temperature, which was about 42% and 31% compared to room temperature, respectively for CFRP and GFRP. Another study by Ahmed et al. [60] highlighted that higher strain rates result in increased tensile strength of SFRP.

In order to increase the flexural strength of plain concrete and RC elements, FRP plates, strips or fabrics can be used externally [7]. The length of the SFRP, binder content, heat treatments, and pre-activation of fibers significantly influence the flexural and tensile strength of members enhanced with SFRP [64]. Jin et al. [65] studied the splitting tensile strength of the chopped basalt fiber rein-

Table 1
Search string of this study.

Year	Sources	Search string
2012–2022	Scopus (Title, Abstract, Keywords) and	("Fiber reinforced polymer" OR "FRP") AND ("Synthetic" OR "Hybrid" OR "Natural") AND "Concrete" AND ("Structural application" OR "Infrastructure" OR "Performance") AND NOT ("Fiber reinforced concrete")
2012–2022	Web of Science (All Fields)	("Fiber reinforced polymer" OR "FRP") AND ("Synthetic" OR "Hybrid" OR "Natural") AND "Concrete" AND ("Structural application" OR "Infrastructure" OR "Performance") NOT ("Fiber reinforced concrete")

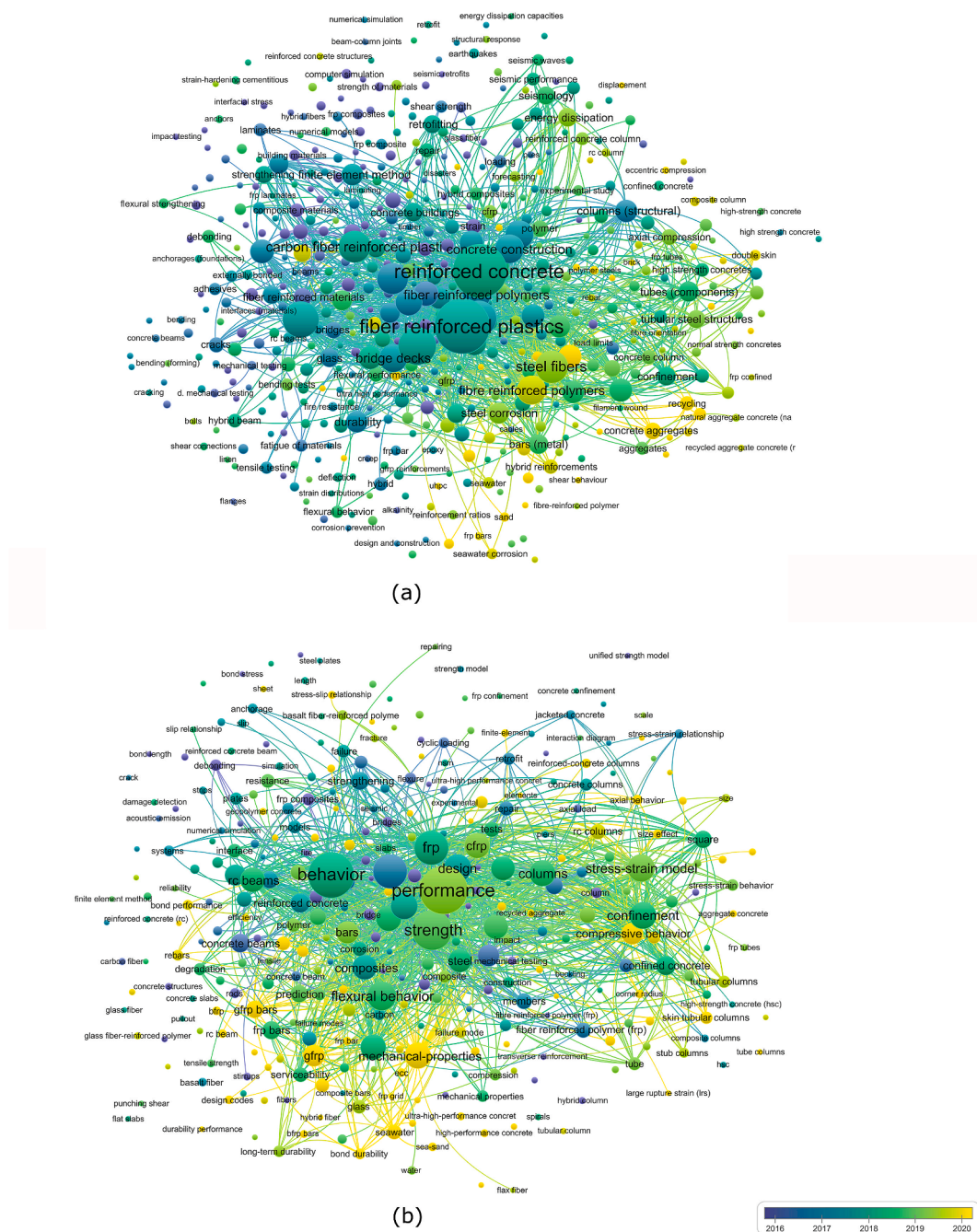


Fig. 3. Network visualization based on keyword co-occurrence analysis: a) Scopus data; b) WoS data.

forced concrete and found that pretreated fibers give higher strength than the normal fibers in terms of the split tensile strength of concrete [65]. The performance of flexural strengthening of concrete, steel, and timber using SFRP is detailed in [Table A1 in Appendix](#). The EB or NSM method can be used to improve the flexural strength of members. [Table A1](#) shows that the performance of NSM strengthened beams under bending has been superior to that of the widely used EB method [66]. Yao et al. [67] also experimentally proved that wrapping a column with an EB-FRP sheet has no impact on increasing flexural performance during bending. However, the NSM strip strengthening technique has been proven to be beneficial. Furthermore, comparing CFRP and GFRP flexural strengthening configurations, it was discovered that a strengthening composite material of GFRP has a strong elongation at rupture, improving the ductility [68].

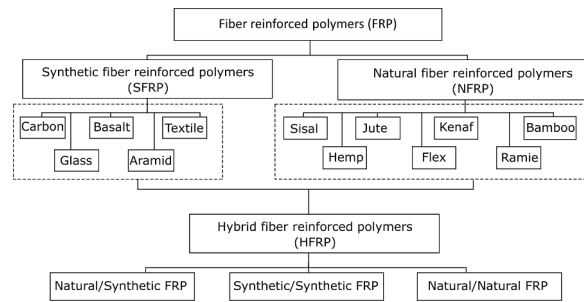


Fig. 4. Commonly used fibers in FRP composites for infrastructures applications.



Fig. 5. Different type of Synthetic fiber reinforcement bars [50].



Fig. 6. Different types of discrete synthetic fibers used in the construction industry [14,15].

3.1.2. Bond behaviour of SFRPs

The bonding performance of SFRPs is a major factor affecting the SFRP composites' strength. Many researchers have utilised various test methods to investigate the bond nature of SFRP and concrete [69,70]. The single shear test, double shear test, direct pullout and bending beam test methods (Fig. 7) are the most common method to assess the bond performance of SFRPs.

The bond performance of CFRP strips bonded to concrete blocks is measured through the maximum load carried by the interfacial area of the bond before failure in a pullout test [35]. A study by Sen et al. [76] highlights that the width of the bonded region and the thickness of the FRP repair are the main factors influencing the failure load of the FRP-concrete system. Further, increasing width and thickness have resulted in increased failure load. The bonding between FRP and concrete is suffered from brittle failure, and the energy released at the failure of the bond is sufficient to create the failures in the FRP [35]. Nepomuceno et al. [72] stated that the bond strength FRP depends on the fiber type, embedded length, SFRP bar diameter and SFRP bar surface treatment. Nepomuceno et al.

Table 2
Mechanical properties of several SFRP [63].

Detail	Tensile strength (GPa)	Elastic Modulus (GPa)
GFRP	2.4–3.5	72–87
CFRP	2.1–5.5	200–500
AFRP	3.1–3.6	58–130
BFRP	4.2–4.8	89–110
GFRP- Epoxy	1050	55000
CFRP- Epoxy	1500	18000
AFRP- Epoxy	1400	76000

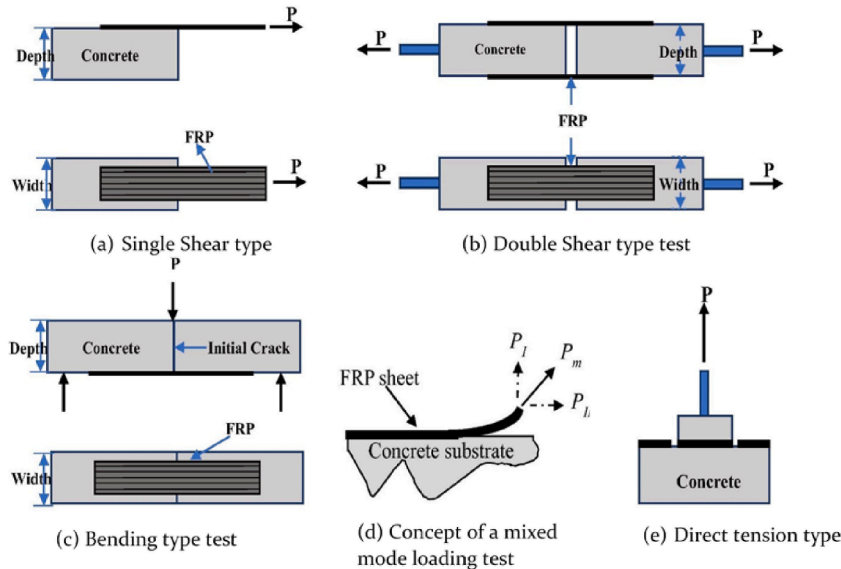


Fig. 7. Different types of FRP-concrete bond test [71].

[72] also found that BFRP bars have better bond strength than GFRP and CFRP bars. Moreover, increasing the bar diameter reduces bond strength, and surface deformations or sand coating can be used to improve bond strength.

The long-term bond performance of different types of SFRPs under the influence of seawater concrete has been investigated [73]. Further, this study evaluates the long-term bond strength between concrete and steel, GFRP, CFRP and BFRP bars [73]. The CFRP bar gives higher bond strength after being immersed in seawater for 30 days compared to other reinforcement. Further, Yang et al. [74] have concluded that the bond-slip behavior between BFRP bar and concrete in seawater is influenced by the strength of the concrete and the size of the BFRP bar. Hassan et al. [75] also investigated the long-term performance of BFRP bars in concrete under aggressive environments. Their study concluded that the highest reduction in the bond strength occurred after six months at 40 °C exposure.

Another study by Ge et al. [76] investigated the bond strength of steel-SFRP composite bars under chloride corrosion. The bond strength ratio of the steel-SFRP composite bar and SFRP rebar to steel rebar ranges from 0.98 to 1.04. Their study showed that steel-SFRP composite bars and SFRP rebars have comparable bond strength to conventional steel rebar. Dong et al. [74] investigated the bond durability of basalt fiber reinforcement polymer (BFRP) bars, steel fiber reinforced polymer bars, steel fiber reinforced composite bars (SFCB) and steel bars in the ocean environment. According to the results related to degradation mechanisms, peeling of the basalt fiber reinforced polymer surface has experienced instead of commonly occurring shear failure. Further, the study concluded interesting facts on bond stress retention where, in BFRP, 92% of retention was observed in the wet-dry cycling environment, and it was 78% when immersed in seawater. Oppositely, in steel bars, no significant bond strength reduction was recorded but pits were formed on the bar surface causing severe corrosion [73]. Another similar study conducted by Dong et al. [77] investigated the bond durability of BFRP, GFRP bars, CFRP and steel bars in seawater under different resin types, fiber types, immersing durations and temperatures. Results show that basalt-epoxy bar to concrete exhibited no change in bond strength while basalt-vinyl ester and glass vinyl ester were degraded. Moreover, in terms of the long-term durability such as after 50 years' service life, the study concluded that the bond strength retention of basalt -vinyl ester will remain above 93% in a dry environment, 78% in a moist environment and 47% in a saturated moisture environment. Furthermore, Mukhtar et al. [78] carried out experiments to study the FRP concrete bond performance of concrete made with and without steel slag under various hygrothermal conditions and found that FRP-concrete bond strength increased as temperature increased at low humidity (dry heat).

Altamas et al. [70] study suggested that higher adhesion and bond strength can be achieved when using sand-coated BFRP compared to ribbed GFRP bars. Furthermore, Dong et al. [77] discovered that ribbed BFRP bars with surface sand-coating reduced short-term bond strength while considerably improving bond durability. Moreover, the ribbed SFRP bars had greater short-term bond

strengths than the steel bars. Lorenzis et al. [79] investigated the mechanics of the bond between NSM FRP rods and concrete. Their study conducted the experimental test to evaluate the effect of groove-filling material, bonded length, groove size and surface configuration of the rod on the bond performance. The average bond stress related to ultimate adhesion and residual friction at the epoxy-concrete interface is varied with bonded length and groove size. The mortar-filled grooves do not show satisfactory results due to the low strength of mortar compared to epoxy. Furthermore, the bond's ultimate load increases when the depth of the groove increases. Ahmad et al. [80] proposed a new method for increasing the bond strength of very smooth CFRP bars by applying an absorbing/protecting sheet (peel ply) to their surface, which is the most cost-effective option.

3.1.3. Fire performance of SFRPs

The SFRP strengthened concrete structures require protection to sustain their strength to prevent combustion of the material itself and resin of SFRP under fire. The glass transition temperature (T_g) is defined as the threshold at which properties of FRP materials start to change significantly [81]. The polymer matrix, initially with a rigid and brittle structure, becomes soft and rubbery when the temperature increases. This will reduce the strength properties and create structural failures in the composite structure. Venkatesh et al. [82] specified that the T_g of SFRP composites lies in the range of 65–120 °C. A study by Bazli et al. [83] highlights that the degradation of mechanical properties of SFRP composites could occur before it reaches T_g . Furthermore, at elevated temperatures (300–400 °C) the organic matrix and the fibers used to reinforce polymer composites start to decompose and release heat, smoke and other toxic gases [83]. This action induced deterioration of the matrix and fibers, reducing the physical and mechanical properties of SFRPs [81].

Generally, two types of test procedures, namely steady state and transition state conditions used to evaluate the fire performance of SFRPs. In the steady state, the specimens are heated to a predefined temperature and then subjected to load up to failure. Whilst in the transition state, the specimens are loaded to a fraction of their ambient temperature strength and then heated to failure. Wang et al. [84] conducted 57 tension tests for different SFRP composites at various temperatures to determine the temperature at which an SFRP material retains just 50% of its initial room temperature strength. Their study found that CFRP and GFRP lose 50% of their strength at elevated temperatures of 250 °C and 325 °C, respectively. Furthermore, the strength and stiffness of both GFRP and CFRP significantly reduced when they were exposed to over 400 °C.

Cao et al. [85] also performed a series of tests to evaluate the tensile strength of CFRP and hybrid (carbon/glass) FRP sheets at elevated temperatures (maximum 200 °C) under steady-state conditions. The experimental tests found that the tensile strength reduces significantly at temperatures below 55 °C (which is taken as the approximate T_g temperature for the system) and has remained constant up to 200 °C. Similar trends have been found in the tensile tests conducted by Wang et al. [86] and Chowdhury et al. [87] for both steady and transient states. Furthermore, Wang et al. [86] found that the CFRP plates exhibits two considerable reductions in tensile strength. The one is in the temperature range of 20–150 °C, reducing the room temperature strength by about 60%. The second one is in the temperature range of 450–700 °C, which reduced the room temperature strength by about 96%. The ultimate strength of the tested specimens was decreased to approximately 50% of room-temperature strength at 300 °C. Tests conducted for specimens of GFRP uni-directional sheets under a steady state by Chowdhury et al. [87] showed that tensile strength reduction occurred at 60 °C (i.e. 15 °C lower than T_g).

Another study by Hajiloo et al. [88] experimentally studied the behavior of three different type of GFRP reinforcing bars at high temperatures. The first GFRP bar type was coated with sand. The second one was made with helical braid of fibers around the bars and sand coat. Third one was containing ribbed surface that produced from pultrusion process of hardened bar, which ensured that there is no influence from this rib to the strength of GFRP bar. It was observed that at temperatures below 250 °C, the first type of GFRP bar retained 80% of its strength. Further, the strength was dropped dramatically about 35% at temperatures of about 400 °C. The strength was degraded linearly up to 250 °C in the second type GFRP bar. Then, the strength was retained unchanged up to 350 °C. The residual strength of third type bars was 51 and 36% when it exposed to 438 °C and 486 °C, respectively.

Previous research studies suggested several ways to avoid fire impacts on various SFRPs, notably the resin exceeding T_g under fire (in Appendix Table A2). It was amply proved that the usage of cement-based adhesive with various materials could be a sustainable method to avoid resin exceeding T_g .

3.1.4. Durability of SFRPs

The durability performance of SFRP materials used to strengthen modern concrete plays a vital role in preserving the structural integrity of infrastructure. The corrosion of the steel bars used in the concrete reinforcements is the major issue that easily shortens the life of the concrete structure. This induced corrosion creates material and strength deterioration, further increased by more aggressive prevailing conditions. Since SFRP is a non-metallic material, it has great potential to overcome RC's material and strength deterioration. This is because SFRP does not appear to be affected by chlorides and carbonization. This indicates that SFRP help to maintain the pH of concrete compared to that of from RC. Furthermore, SFRP can prolong its life when it is employed with pozzolanic materials. This is because it reduces the alkalinity of the concrete. The use of SFRP in marine and offshore environments is rapidly increasing [89]. As a result, it is very important to examine the deterioration of SFRPs in such aggressive environments. Many tests were conducted on this aspect to evaluate the performance of FRPs with exposure to different environmental conditions [69,90].

A recent study by Kootsookos et al. [91] experimentally investigated the effect of seawater immersion on the durability of GFRP and CFRP composites. Their study discovered that GFRP absorbs more moisture than CFRP composites. This is because of the emulsion size employed on glass fibers, which enables higher water absorption at the fiber/matrix interphase than the silane size on carbon fibers. Furthermore, Sen et al. [92] discovered that the presence of the Si–O–Si group in GFRP composites was more sensitive to the OH⁻ ions in the alkaline atmosphere in terms of the strength degradation as it induced the deterioration. This indicates that GFRP

composites have a low resistance to moisture, alkali, and acids. Wei et al. [93] also examined the degradation of basalt fiber and glass fiber/epoxy resin composites in seawater. The change in mass gain of BFRP and GFRP composites after being immersed in saltwater is caused by a combination of two effects that are water absorption and soluble material extraction. The tensile and bending strengths were reduced with the increasing immersion time. This highlights that the material has suffered physical damage and/or irreversible chemical degradation.

A study by Halim et al. [94] identified three main mechanisms that were arising durability issues of SFRP composites which are chemical or physical changes or micro cracks in the polymer matrix, loss of adhesion or detachment at the fiber/matrix interface, and reduction in strength and elastic modulus of SFRP composites. Furthermore, a study by Bank et al. [95] found that water molecules in the resin degraded the strength and elastic modulus of SFRP composites. As it acts as resin plasticizer, thus breaking the van der Waals bonds in polymer chains. This also can create deformation that leads to fracture and toughness. The swelling stress can cause permanent cracking of the polymer matrix, hydrolysis and adhesion of the fiber matrix [96].

Moreover, Rifai et al. [97] studied the durability of BFRP reinforcing bars in alkaline solution and damp concrete conditions for three, six, and nine months. Results show that the tensile strengths of specimens conditioned in alkaline solution at 20 °C for three, six and nine months were about 4, 15 and 25% higher than that obtained at 60 °C. In contrast, raising the temperature of the specimens conditioned in moist concrete from 20 to 60 °C resulted in an additional 6.5% tensile strength loss regardless of the conditioning duration. The moisture content of BFRP specimens conditioned in the alkaline solution is higher than their counterparts encased in damp concrete. Duo et al. [52] also studied the durability of GFRP and BFRP bars in water, acid, salt, and alkali solutions. The strength of GFRP and BFRP bars is significantly affected under an alkaline environment at lower temperatures such as 20 °C and 40 °C [52]. Furthermore, in the alkali, acid, and salt solutions, GFRP bars demonstrated greater corrosion resistance than BFRP bars. It also concluded that BFRP bars degrade more quickly in the same environment than GFRP bars [52].

Overall, the factors affecting the alkaline attack rate of FRPs have been determined as the susceptibility of the fibers to an alkaline environment. Primarily, the alkaline diffusivity of the resin and the degree of protection allowed for the fiber to play a major role in how the concrete acts in an alkaline environment. Also, the quality and strength properties of the fiber-resin bond are vital. Otherwise, alkalis can penetrate and attack fibers causing sudden deterioration. Furthermore, several other factors influence the alkaline nature of FRP such as temperature, influence rate and rate of reaction in diffusion, alkalinity concentration based on the cement type and mobility of alkaline ions, which are affected by the degree of saturation. The use of appropriate SFRP can significantly increase the life of the structure and minimize maintenance requirements.

3.2. Natural fiber reinforced polymers (NFRPs)

Natural fibers are not manmade fibers and can be produced from plants, which are non-synthetic [35]. The changing regulations on the recyclability of composites and environmental concerns compelled researchers to create materials from renewable sources. Flax, hemp, sisal, hemp, Kenaf and jute are examples of natural fibers that enable the required sustainability [98,99]. The fibers shown in Fig. 8 are fully processed and ready to use to develop natural fiber-reinforced composites for infrastructure building construction. This composite is lightweight and less expensive than that made from synthetic fiber reinforced [100]. Further, these natural fiber composites have the potential to mitigate the issue related to thermal and acoustic when used as insulating materials.

Thus, infrastructure would gain a long service life when they employ natural fiber composites in their construction and building materials. Furthermore, it will decrease construction waste that sends to landfills, boost energy efficiency and promote the notion of sustainability [101,102].

Natural fiber composites outperformed in industrial applications compared to glass fiber-reinforced composites [103]. Flax fiber is a strong candidate to replace the glass as a composite reinforcing material due to its nature, such as low cost, lightweight, and high strength/stiffness for structural applications [101,104]. In many applications, natural fibers are utilised as bundles that are still held



Fig. 8. Natural fibers: (a) Jute; (b) Hemp; (c) Kenaf; (d) Flax; (e) Ramie; (f) Sisal; and (g) Bamboo [99].

together by pectin. These bundles need less processing, and their strength capability is lower than individual fibers. The reinforcing fibers' length can considerably impact the composite's strength and stiffness, especially in cases when interfacial bonding is weak [105,106].

3.2.1. Strength performance NFRPs

The fiber content of natural fibers is the most important element in determining the performance of NFRP composites [25]. Ku et al. [107] studied the composite performance made with nonwoven mats from hemp and polypropylene fibers mixed together in different proportions. The study revealed that increasing hemp fiber percentage in the above composite composites decreased tensile strength. A maximum of 34% tensile strength reduction was observed when using 70% hemp fibers in NFRP composites in the perpendicular direction. The tensile strength of NFRP composites with fibers oriented perpendicularly was found to be 20–40% lower than that of NFRP composites with fibers oriented parallelly [107]. The elastic modulus and tensile strength of NFRP rise with an increase in the fiber weight ratio. Further, the fiber weight ratio falls below the optimal level, and the load is transferred to more fibers that are well bound to the resin matrix. This was the reason for resulting in the improved tensile characteristics of NFRP [108,109].

The tensile strength of NFRP composites made with kenaf, flax, and jute increased with loading up to a maximum or optimum value (i.e. the value before dropping back). A progressive increment is observed for natural fibers used in the study such as rice hull, flax and various mesh sizes of hardwood, when it comes to the variation in Young's Modulus of NFRPs. A study by Ku et al. [107] shows that the NFRPs bleached with hemp fiber have resulted in the initial increase of tensile strength to 30 MPa from 20 MPa at 5% fiber content. Further, when increasing the fiber replacement level to 20%, the tensile strength was dropped to a low 23 MPa. However, the tensile strength increased again to a value closer to 5% fiber content when the fiber content was 30% [30].

Borri et al. [110] used flax, bamboo, basalt, and hemp fibers strips to reinforce the tension zone of the timber beam. Their study found that natural fiber reinforcement increases timber beams' load-carrying capacity, strength and stiffness. Reinforcing low-quality timber beams using basalt fibers resulted in excellent strength and ductile behavior. Flax fiber reinforcement yielded much greater improvements in capacity than basalt reinforcement. A recent study by Moezzi pour et al. [111] assessed the effectiveness of reinforced plywood with natural fibers. Their study found that using kenaf and palm fibers to improve the practical characteristics of plywood is beneficial. Further, Kenaf fibers had better performance for reinforcement of plywood than palm fibers. Carvalho et al. [112] study assessed the technique to reinforce the flawless timber beams using sisal fibers woven in the direction perpendicular to the length of timber. The addition of sisal fiber showed a reasonable increase in the flexural strength modulus of flawless timber.

A similar way of using sisal fibers to strengthen low-mechanical resistance wood species has been experimented by Mascia et al. [113]. Reinforcing the timber beam with sisal fibers increased the values of stiffness properties and modified the brittle mode of failure to ductile nature. The sisal fibers reinforcement also helps to avoid the fragility of critical tensile regions of the beams. Wang et al. [114] also investigated the structural behavior of externally strengthened timber beams with flax and basalt fibers. Results show that the flexural capacity of beam strengthened with basalt FRP is lower than that of flax FRP. It was also found that the failure modes of FRP strengthened beams changed from tensile failure to FRP debonding as their maximum bending load increased. Khadafi et al. [115] investigated the flexural capacity of glulam timber beams reinforced by bamboo strips. Results revealed that strengthening the tension side of the beam using bamboo strip reinforcement resulted in a substantial increase in flexural characteristics compared to the beam without reinforcement.

Recent study by Valdes et al. [116] provides an overview of how cross-laminated timber (CLT) and natural fibers may be used to create high-performance structural components. Their study successfully introduced the prospect of reinforcement on existing CLT slabs with external flax fibers layer in the tension zone of the specimens. A significant increase in flexural capacity, stiffness, and midspan deflection was reported in three-layered CLT panels. However, in five-layer CLT panels, the reinforcing efficacy was insignificant. The different strengthening methods of timber elements using natural fibers are shown in Table A3 in Appendix.

3.2.2. Bond behavior of NFRPs

The bond between the NFRP composite with the structural members is an important factor in the composite performance of NFRP-reinforced structural members under load. Hallonet et al. [117] used a double shear test to investigate the bond behavior of flax FRP composites bonded to a concrete substrate. The tensile load was applied on the concrete blocks, which were bonded by two parallel bonded FRP strips symmetrically [117]. It was observed that the double lap shear tests generate concrete failure in all specimens. This indicates higher bond strength can be obtained between the FRP and the concrete. Flax FRP exhibit a shear failure of the concrete. Furthermore, the composite rupture was observed, indicating that the concrete is the weak element. This clearly supports flax FRP's superior bond performance with concrete.

In contrast, Vinh et al. [118] observed a sudden and explosive debonding of sisal FRP in the sisal strengthened RC beams. An end-anchorage system was developed to avoid sisal FRP debonding (Fig. 9). The anchorage system is made up of full threaded bolts and steel plates. This proposed anchoring system was proven highly efficient in preventing sisal FRP delamination from the concrete substrate.

Ramakrishnan et al. [119] investigated the bond behavior of flax FRP with steel using a single lap shear test. For the bonding between the flax FRP and the steel layer, two adhesives were used, which are commercial epoxy adhesive and natural rubber. The force-displacement curves from the single lap shear test revealed that the joint with flax composite and epoxy adhesive was stronger, but the bonding with rubber was weak. Moreover, the adhesive residue was found on the substrates, indicating that the epoxy bonded connection failed due to a combination of adhesive and cohesive failure. This indicates that the composite and the steel layer are well adhered.

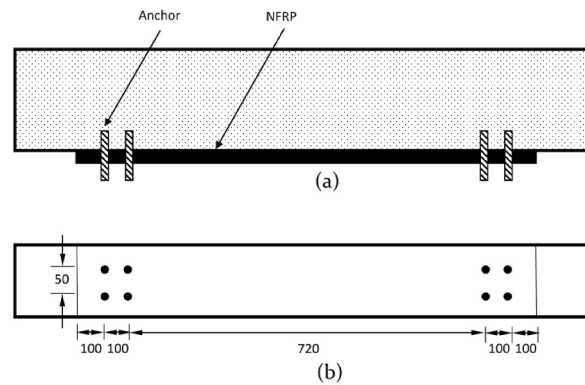


Fig. 9. Proposed anchorage details for sisal FRP strengthened RC, (a) sectional view of the arrangement and, (b) plan view of the arrangement (all dimensions are in mm) beams [118].

3.2.3. Fire performance of NFRPs

Natural fibers have several advantages, but one of the most significant disadvantages is their low fire resistance. The decomposition happens at temperatures ranging from 300 to 500 °C, producing flammable gas, char, and smoke. The presence of cellulose in natural fibers and hydrocarbon in polymers contributes to the high flammability nature of natural fiber-reinforced composites [120]. Lignin-rich fibers such as flax and hemp have been discovered to have lower heat release rates [121]. Thermal degradation of hemp fibers is a one-stage process that begins at about 150 °C and is related to the breakdown of hemicellulose and lignin [122]. When comparing flax fiber samples to glass fiber counterparts, Chai et al. [123] show that the flax fiber samples burn earlier, emit more heat overall, and take longer to extinguish. During the combustion process, the flax fiber samples showed substantial distortion, whilst the glass fiber samples remained intact.

Flame retardant treatments are necessary to broaden the range of uses for NFRP [124,125]. Flame retardants help to reduce the flammability of fiber reinforcement, polymeric matrix, or whole composite. Therefore, improving the fire retardancy of the polymeric composite material is important before it applies to infrastructure. Previous studies have indicated that adding fire-retardant additives such as halogen, nitrogen, mineral-based retardants, and phosphorus-based chemicals to NFRP composites can improve their fire resistance [98,124]. Bachtiar et al. [126] examined the impact of varying mass concentrations of ammonium polyphosphate and aluminium hydroxide as fire retardants on flax FRP composites. Thermal stability tests indicated that employing ammonium polyphosphate and aluminum hydroxide as fire retardants in composite treatments significantly increased thermal resistance [126]. However, flame retardant's effectiveness depends on the matrix type and the flame retardant utilised. Guanylurea methylphosphona, an organophosphonate flame retardant with greater nitrogen and phosphorus concentration, was the most effective flame retardant for flax-based FRP composite [98]. However, more research is needed to understand better the mechanisms and behaviors that influence the characteristics and appropriate selection of flame retardants for NFRP composites.

3.2.4. Durability of NFRPs

The capability of NFRP to resist cracking, chemical degradation, delamination, and wear for a defined amount of time under specified load conditions and environmental conditions is referred to as durability [127]. Sen et al. [127] studied the moisture absorption of concrete composites made with sisal, jute, carbon and glass when exposed to wet environments. The composites were first weighed and then heated with normal water until bubbling for 30 min. Afterwards, the composites were removed and wiped with a cloth to measure the moisture content and thickness swelling. Results showed that the moisture content percentages for sisal and jute FRP were 5.9% and 6.6%. It was comparatively higher than that recorded for carbon and glass, 1.4% and 0.6%, respectively. Thickness swelling is an essential characteristic that indicates the composite's durability performance [127,128]. Water absorption is reduced when the thickness swelling is reduced. The thickness swelling for sisal was 2.7%, 8.9% for jute FRP and 7% for carbon and glass composites [127]. Further, the thickness swelling index of the Sisal FRP composite was the lowest [127]. The tensile strength of both FRP composites experienced an increment in the wet conditions compared to the dry conditions, whereas the tensile strength increased by 6% for sisal and 8% for jute, while it was 1.3% and 1% for carbon and glass, respectively [127].

Yan et al. [100] examined the impact of ageing flax FRP composites in water, seawater, and alkaline (5% NaOH) solutions in civil engineering applications. After prolonged immersion in water, salt water, and 5% NaOH solution, all of the composites faded, and the color of specimens degraded. The change in the color of specimens indicates that the chemical changes in the epoxy matrix occurred due to matrix's heat oxidation. This color change might be due to photo oxidation, which helps in degrading the polymer surface. During immersion, the absorbed water molecules in the epoxy matrix accelerate photo-oxidation processes due to the availability of OH⁻ and H⁺ ions throughout the weathering cycles.

Furthermore, the weight increase is an essential metric that may be used to assess the flax FRP susceptibility to deterioration and forecast its long-term durability. The weight gain for the specimens immersed in water, seawater and NaOH solution at two months is 8.56%, 7.98% and 9.01%, respectively. It was found that flax FRP composites accumulate more weight than SFRP composites. Moreover, the specimens exposed to different atmospheres reduced tensile strength. Tensile strength was reduced the most in a 5% NaOH solution [100]. It shows that the presence of alkaline salts in the NaOH solution causes quicker tensile strength deterioration. The de-

terioration of flax fibers, the epoxy matrix, and the link at the fiber/matrix interfaces during immersion is found to be important for the loss in tensile characteristics of flax FRP composites. The degree of degradation is strongly influenced by the immersion temperature and the type of polymer matrix.

Cheour et al. [129] investigated the mechanical and damping properties of flax FRP composite underwater ageing. Their study found that fiber orientation significantly influenced water absorption and bending strength. The changes in the bending moduli of aged flax FRP composite are largely irreversible after drying, indicating that water ageing caused permanent degradation of the mechanical characteristics of flax FRP composite. Further, Hristozov et al. [130] conducted accelerated durability tests to compare the long-term mechanical performance of flax FRP composite exposed to different types of atmosphere. Flax FRP composite specimens retained strength of 88%, 71%, and 68% in water at 20 °C, 50 °C, and 60 °C, respectively after 125 days of conditioning. Whilst it was about 81%, 69%, and 65% in salt water, and 74%, 59%, and 49% in alkaline at 20 °C, 50 °C, and 60 °C, respectively. The flax specimens showed about 23% higher strength retention than the GFRP after 125 days of conditioning in water, salt water and alkaline [129]. This demonstrates that composites comprised of flax fibers can enable long-term mechanical performance than the GFRP.

The use of NFRP in infrastructure enables significant environmental and long-term advantages. Sisal FRP, jute FRP, and flax have high confinement strength and confinement modulus, making them ideal ductile materials with excellent mechanical characteristics [107]. Their use in various infrastructure applications would undoubtedly aid the efforts to achieve a more sustainable, greener environment with improved rural development that promotes biodiversity.

3.3. Hybrid FRPs (HFRPs)

The HFRP composite materials are created by combining two or more different fibers in a single matrix (Fig. 10). The hybrid fiber can be made from combinations such as natural–natural fibers, natural–synthetic fibers and synthetic–synthetic fibers in a single matrix [36]. Kenaf/Glass [131], Jute/Glass [132], Carbon/glass [133], Basalt/Hemp [134], Flax, Hemp and Jute [135], Aramid/Sisal [136] are few examples of hybrid fiber reinforced polymers currently in practice. Hybrid composites have a wide range of characteristics needed to meet various required infrastructural applications. The reinforcing of multiple fibers of varying diameter and length provides improved material and mechanical characteristics (i.e. strength, durability, low weight and density) than that of from single polymer matrix composite [137]. Further, to decrease the environmental footprints of infrastructure, natural hybrid composites can be used instead of SFRP composites [31,35].

3.3.1. Mechanical and durability performance of HFRPs

The HFRP composites are one of the developing technologies that are gaining momentum for use in a variety of infrastructure construction [32]. Recent study by Dhinakaran et al. [137] found that the hybrid composite material performed better and had higher strength in tensile, flexural and hardness tests than the other composite materials. Furthermore, it was discovered that hybrid fibre composites' tensile and flexural strengths improve as the fiber volume percentage increases. Another study by Asim et al. [139] developed an HFRP composite by combining pineapple leaf fiber and kenaf fiber and incorporating them into a phenolic matrix. The mechanical characteristics such as tensile, flexural, and impact strengths of this HFRP composite increased significantly. The inclusion of kenaf fibers to composites up to 70% by weight resulted in improved tensile and flexural strengths of HFRP composite. A summary of the mechanical and durability properties of hybrid fiber-reinforced polymer composites is detailed in Table 3. This table provides an insight into the HFRPs performs in terms of tensile, flexural strength and durability.

3.4. Environmental footprints of FRPs

The FRP composites benefit to modern industries in many ways due to their numerous advantages, yet they can generate many of environmental issues. The manufacturing of SFRPs creates substantial greenhouse gas emissions and natural resource depletion [143]. Carbon fiber manufacturing creates roughly 15 times more CO₂ on a weight basis than traditional steel [144]. Furthermore, the carbon fiber production is 4.6 times more energy-intensive (i.e. 230 MJ/kg) than traditional steel (i.e. 50 MJ/kg) [145]. However, carbon fiber recycling may reduce its energy intensity compared to steel [146]. In contrast, the manufacturing of 1 kg glass fibers consumed 54.7 MJ, which is less than that of from carbon fiber production [147]. However, even if these composite materials have a significant environmental impact, they remain a viable alternative to steel reinforcement. As it enables the required strength while reducing adverse environmental impacts. A study by Maxineasa et al. [148] highlighted that the steel RC beam strengthened with CFRP increases the load capacity and reduces adverse environmental impacts.

The environmental impacts of an RC beam are influenced by the amount of cement used in the concrete mix as well as steel reinforcements utilised in the beam. The use of these innovative FRP fibers for strengthening and retrofitting or upgrading existing/old concrete structures can reduce the cement contents and amount of steel reinforcements. This will enables adequately meeting the re-

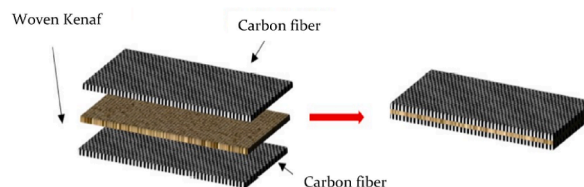


Fig. 10. Hybridization of a woven kenaf with synthetic fibers [132,138].

Table 3
Summary of mechanical and durability properties of HFRP composites.

References	Fiber arrangement in HFRP	Mechanical and durability performance
Khan et al. [140]	Jute-Kenaf-Jute and Kenaf-Jute-Kenaf FRP	1. Improved the tensile strength and flexural strength 2. Kenaf-Jute-Kenaf shows better tensile strength and flexural strength than Jute-Kenaf-Jute FRP 3. Kenaf-Jute-Kenaf FRP had the best structural properties with less fiber breakage
Yorseng et al. [141]	Kenaf-Kenaf-Kenaf, Sisal-Sisal-Sisal, Kenaf-Sisal-Kenaf, and Sisal-Kenaf-Sisal FRP	1. Better weathering resistance 2. All the composites are stable up to 295 °C
Aisyah et al. [138]	Carbon-Kenaf-Carbon FRP	1. Lower tensile strength than CFRP 2. Highest flexural modulus 3. Lower impact strength
Romanzini et al. [32]	Glass/ramie FRP	1. Low weight and higher water absorption 2. Improved impact and interlaminar shear strength
Dhakal et al. [142]	Carbon/flax FRP	1. Water absorption is improved when compared to before hybridization 2. Tensile and flexural strengths were found to decrease with an increase in percentage moisture uptake
Atiqah et al. [33]	Kenaf/glass FRP	1. Higher value of flexural, tensile and impact strength 2. Mercerization procedure improved the adhesion between the surface of the fiber and the matrix

quired level of structural safety and reduce the environmental footprint by reducing the need to construct new buildings. This will also contribute to reducing the negative environmental impacts produced by the fabrication stage. Dai et al. [149] show that replacing conventional prestressed concrete with FRP composites in pedestrian bridges reduces CO₂ emissions (approximately 26%). During the material manufacturing and construction periods, CO₂ emissions can be reduced by 18% and 70%, respectively. Natural fibers such as flax, hemp, jute, and kenaf, have a smaller environmental impact than carbon and glass fiber [150–152].

Jute fiber has a greater carbon footprint than other natural fibers as its production per hectare is lower [153]. Natural fiber composites have a carbon footprint of around half that of glass fiber composites, including processing steps (Fig. 11). Joshi et al. [147] and Broeren et al. [151] analyzed the life cycle assessment of natural fibers over glass fibers. Their study stated that natural fiber manufacturing has a lesser environmental effect than glass fiber production, and the end-of-life incineration of natural fibers saves energy and reduces greenhouse gas emissions. Malviya et al. [154] demonstrated in their case study that natural fiber composites have emerged as a viable alternative to synthetic fiber reinforced composites. The cradle-to-gate environmental impact comparisons of natural fiber textile reinforced bio-based composites also show a reduction in greenhouse gas emissions [155].

Overall, the use of FRP in the form of synthetic fiber and natural fiber in the building sector has been shown to reduce environmental impacts with the inclusion of recycling for synthetic fibers. Further, natural fibers are greener construction materials for sustainable construction. Therefore, the FRP materials are considered to be a vital component in the sustainable growth of the building sector and should be appropriately marketed as an environmental solution in the present global context [156].

4. Applications of SFRPs in infrastructures

Generally, SFRPs are used to reinforce the structural elements in the buildings such as beams, slabs, and columns in the forms of sheets, laminates, and bars. However, limited application in the infrastructures, especially ships and offshore platforms, bridges, aircraft parking and runways, pipes, pavements, slope stabilizations, dams and elevated decks [157]. The following subsection discusses a few worldwide applications of SFRPs in infrastructure attributed to tunnel engineering, electrical transmission lines, bridge structures, nuclear structures and seismic resistance improvements.

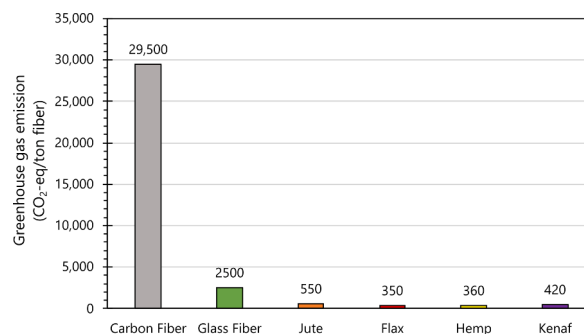


Fig. 11. Comparison of the greenhouse gas emissions per ton natural fibers [153].

4.1. Tunnel engineering

The safety of tunnel operations is significantly suffered from damages induced by high earth pressure created from material deterioration, water leakage, geological issues and frost damage [158,159]. Therefore, to preserve tunnel structure safety, progress must be made on inspection technology, diagnostics and repair technology [159]. The degraded lining concrete of tunnel constructions is very difficult to remove, and it should be strengthened with certain additional materials [158]. The SFRPs can potentially strengthen degraded lining concrete due to their nature such as low weight, simple to handle, quick application, high strength, immunity to corrosion and little occupancy interruption [20].

A study by Jiang et al. [158] assessed the effectiveness of SFRP grids embedded in polymer cement mortar shotcrete on degraded tunnels. This approach proved that the axial stress of a tunnel lining can be effectively reduced. The SFRP grids not only have a high level of durability, but they are also light in weight, high strength and simple to manufacture. Tunnel reinforcement engineering has utilised SFRP grids extensively to control adjacent rock deformation effectively. Furthermore, Mohamed et al. [160] described three field applications of SFRP-RC soft-eyes in tunnel construction in Canada. Their study specified the design methodology, construction methods and required material properties for the SFRP reinforcement in tunnel construction.

Steel bar corrosion is another major problem in tunnel construction, which reduces the life of the tunnel. Mohammed et al. [160] study suggested using GFRP bars as a solution for the corrosion-related issue. The GFRP is high corrosion resistance material, and the weight of a GFRP bar is just a fourth of that of a steel bar with the same dimensions [161]. Due to these advantages, GFRP reinforcing bars are effectively used in tunnel constructions in Canada (e.g. South tunnels – Keele Station, Eglinton Crosstown LRT - Toronto, and the Coxwell sewer shaft projects). This indicates the significant progress in this SFRP application in tunnel engineering [160,161].

Moreover, Caratelli et al. [162] discussed the possibility of substituting the traditional steel reinforcement with GFRP bars in pre-cast segmental lining tunnels. The durability of steel cages is a major concern in tunnels subjected to aggressive environmental conditions. Caratelli et al. [162] conducted several full-scale tests to prove the significant benefits of employing a GFRP cage in tunnel segmental lining construction instead of the steel cage. The large-scale test shows that the use of GFRP instead of steel resulted in the same flexural structural behavior, due to its brittleness, but it exhibits high strength and adequate durability. Further, Caratelli et al. [162] indicate that the greater cost of the GFRP cage compared to standard steel cages may be balanced when the entire tunnel construction and maintenance expenses are considered.

Meda et al. [163] investigated the potential of using fiber-reinforced concrete tunnel segments with GFRP reinforcement in a hybrid approach. A compact ring shape characterises the longitudinal GFRP reinforcement. This strategy can be beneficial in areas of the tunnel (cross-passage or shallow tunnel). This is because the fiber-reinforced concrete alone cannot meet the requirement owing to bending moment action related to the axial force. The inclusion of GFRP bars can be statically beneficial and aid in the resolution of durability issues.

A study by Hara et al. [164] proposed the usage of SFRP corrugate sheet for tunnel lining. SFRP corrugate sheets exhibits superior performance and stiffness compared to steel. The SFRP corrugate sheet has a transparent nature, thus cracks and small movement of rocks can easily be monitored. On the other hand, Yang et al. [12] suggested that carbon fiber reinforcement can be applied to increase the carrying capacity of the tunnel, provide headroom, and allow for normal usage and construction. Moreover, the FRP bolts were utilised as a probe in a 2.5 km long railway tunnel in China's Pearl River Delta to monitor the internal force of the rocks surrounding a mine-shield tunnel for initial support in conjunction with an anchor-head dynamometer [13]. The usage of FRP bolts instead of steel rods has many advantages such as corrosion resistance in terms of durability concerns, strong tensile and low shear strength in terms of strength and very low heat and thermal conductivity in terms of life cycle energy performance.

In tunnel engineering, no matter which design technique is used, the tunnel's primary structure will unavoidably be in the surrounding rock, which has extremely poor environmental conditions and corrosive issues. Because of its high corrosion resistance, the SFRP material could be an alternative to steel reinforcement in this circumstance. However, there is a potential risk of deformation or cracking in SFRP concrete components due to the low elastic modulus of SFRP. Thus, more research and development are needed to improve the elastic modulus of SFRP.

4.2. In electrical transmission lines

Most countries have used traditional materials such as timber, steel, or concrete for electrical transmission lines for the last decades. These materials have several drawbacks, including difficult disposal and tendency to decay in timber, hefty weight and maintenance problems in concrete, and high cost and risk of conduction in steel. The majority of the lighting pole is constructed of steel, which is susceptible to corrosion. Stainless steel and iron steel are commonly used to combat corrosion concerns, but they are also typically produced by hand, making them expensive [165]. Further, the advantages of an aluminium lighting pole are its small weight, high rigidity and strength, and relatively good corrosion resistance. However, when exposed to certain chemicals, aluminium lighting poles exhibit signs of frailty. Thus, the SFRP composites can be a potential option to achieve the classic aims of transmission structures.

Filament winding, pultrusion, and centrifugal casting are the three main production processes for SFRP transmission line components [37]. Godat et al. [166] studied replacing traditional constructional materials such as steel, wood and concrete with fiber glass in electricity transmission lines. The study examined the structural behavior of various SFRP pultruded components composed of E-glass using full-scale laboratory testing. The investigation was conducted on angular section, circular-section, square-section, rectangular-section, W-section and I-section specimens. According to the experimental data, the angle-section members failed due to a combination of global buckling and bending. The square-section, rectangular-section, and W-section specimens failed due to local buckling. The lateral torsional buckling failure was seen in the I-section specimens. Furthermore, Godat et al. [166] found that the fiber

density was linked to the failure of FRP pultruded portions since the failure occurred where the fiber content was low. Moreover, all of the tested specimens displayed elastic behavior. Godat et al. [166] also examined the feasibility of replacing steel with SFRP in electricity transmission towers. The investigation was carried out on a 10-km-long transmission tower with a voltage of 69 kV electricity transmission. Further in this investigation, the span of transmission towers changed from 100 m to 400 m in increments of 50 m to assess the sustainability. It was concluded through cost analysis that the SFRP pultruded circular section with a 200 m span between frames was found to be the best option for replacing steel in electrical transmission lines.

4.3. In bridge structures

Over the last two decades, SFRPs has been increasingly popular in bridge construction and used for basic deck components, pultruded members, and complete load-bearing bridge structures [167]. Due to its numerous advantages, SFRP composites are used to construct bridges worldwide [168].

In the Netherlands, a large number of SFRP bridges have been constructed over the last 20 years [167]. The main rationale for using SFRP for bridge construction in the Netherlands is because of its low carbon footprint, durability, and corrosion resistance. Footbridge in Harlingen was the first 100% SFRP footbridge constructed in the Netherlands [169]. Kim et al. [170] conducted a survey in the form of questionnaire and follow-up interviews with 46 North American transportation jurisdictions to identify the importance of SFRP in bridges in the USA. It was concluded that more than 80% of respondent agencies had adopted FRP materials in their projects. Most of the companies in the USA use SFRP composites for bridge girders, decks and piers (columns), piles, abutments, underground structures, concrete pavement, drains, and culvert liners in the forms of sheets, laminates, structural shapes, bars, and tendons.

The first composite fiber bridge in Australia in early 2002 has proven that SFRP composites represent feasible bridge structural materials [199]. It was designed as a conventional plank bridge where fiber composites coupled with high tensile strength and low weight features with a high compression capacity. Moreover, the Australian Taromeo Creek Bridge was the first highway bridge built on fiber composite rails and reinforced concrete slab, replacing an old wood bridge [171].

The structural application of FRP in steel bridges is typically based on its high strength-to-weight ratio, high tensile strength and corrosion resistance [40]. One of the most notable examples of FRP used in steel bridges is a steel girder bridge in Guthrie County, Iowa, which was reinforced with post-tensioning CFRP rods [44]. The strengthening system was designed to enhance the bridge's load-bearing capacity. Meanwhile, an I-beam bridge in Pottawattamie County, Iowa, was reinforced with pultruded CFRP. This consists of continuous unidirectional carbon fiber in an epoxy matrix to improve the strength of steel composite beams, particularly developed for flexural strengthening [172]. FRP's use to repair and strengthen structural timber bridges has several advantages [46,47]. It can be used as beam end reinforcement, tension reinforcement perpendicular to the grain, bending reinforcement, shear reinforcement and pre-stressed FRP [46]. Table A4 in the appendix shows the detailed application of SFRP in bridge engineering all over the world.

4.4. In nuclear structures

There is a lot of interest in looking into the use of SFRPs in nuclear power plants [173]. SFRP is subjected to prolonged high temperatures, rapid temperature surges, steam and gamma radiation in these particular applications [173]. A study by Homam et al. [173] reported that usage of GFRP and CFRP does not adversely affect the tensile and bond properties of SFRP as a result of gamma exposure. The quantity of publicly available information on the endurance of SFRPs exposed to nuclear power plant environments is very limited [174]. Therefore, more research is needed to ensure the potential of SFRP in nuclear power plant construction.

4.5. Improve the extreme load resistance of infrastructures

4.5.1. Seismic load

Many old structures that are not intended to resist seismic loads are now outdated because of the introduction of stricter design regulations and requirements [175]. Recent earthquakes also urged repairing and reorganising seismically weak structures to minimize damage and losses [176]. SFRPs are recognized as sustainable seismic retrofit materials for concrete buildings [177]. The high-strength fibers in a polymer matrix of SFRP composites help to transfer the extreme load and protect the fiber from extreme weather [178]. SFRP also help effectively avoid the collapse of concrete covers induced by earthquakes and the corrosion of steel reinforcing rebar [179].

The beam-column joints on structures were the primary elements impacted by moderate to large magnitude earthquakes. Flange-bonded FRPs applied to crucial sections of both beams and columns without causing significant damage to the slab are a feasible strategy for increasing flexural capacity in seismically vulnerable buildings [180]. Mahdavi pour et al. [180] investigated the seismic behavior of ordinary RC buildings retrofitted with externally bonded FRP composites with three different SFRP configurations (Fig. 12). It was determined that the frame with a combination of flange-bonded and wrapped layers (Fig. 12c) had higher collapse resistance capacities. The collapse resistance capacity of retrofitted frames with only FRP layers on the columns was minimal (Fig. 12a). Increasing the confining layers to six, on the other hand, was proven unsuccessful. Wrapping columns (Fig. 12b) demonstrated a greater ductility performance.

Most research studies have documented the seismic performance of SFRP-wrapped concrete columns [181,182]. Ghatte et al. [183] study proved that CFRP jackets and external steel ties are significantly more beneficial than the current retrofitting method for seismic retrofitting in terms of ductility and energy dissipation capability. Two examples of earthquake-damaged constructions are the Juan Pablo II bridge and the pier damage of Moorhouse Avenue over the bridge [184]. To increase the seismic performance of the above-mentioned damaged structures, SFRP was applied as an efficient strengthening material.

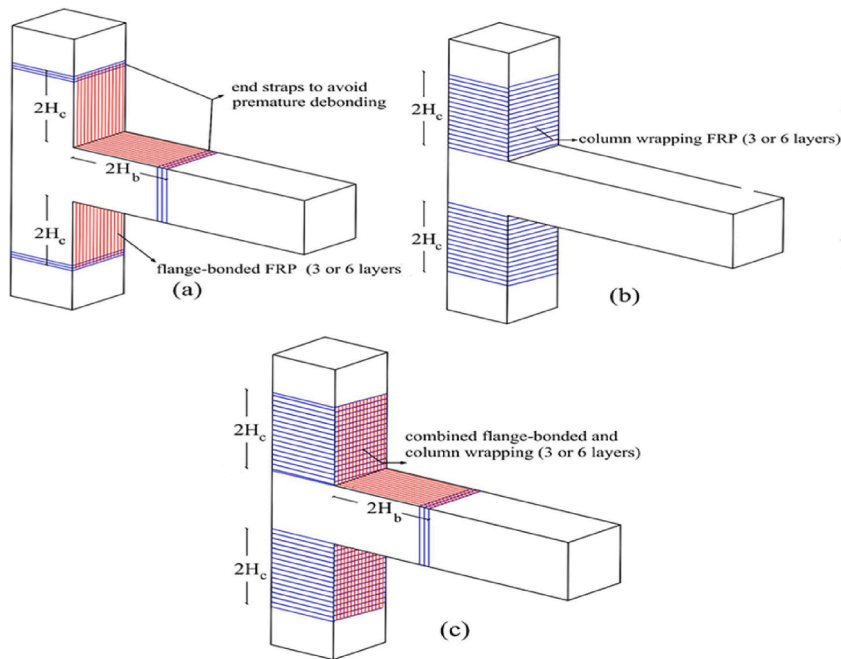


Fig. 12. Different types of retrofitting: (a) flange-bonded FRP with end straps; (b) column wrapping column; (c) wrapping column and flange-bonded FRP [180].

A study by Dai et al. [185] has shown that polyethylene terephthalate (PET) FRP for seismic renewal of square RC columns has a larger tensile capacity than conventional FRPs. Moreover, it significantly enhances the displacement and ductility of RC columns and does not break at the ultimate limit state. Adopting this novel form of SFRP in real-world situations aids in the gradual collapse of columns, which is advantageous in seismic design. Shinozaki et al. [186] suggested the usage of AFRP sheets in strengthening RC beams for seismic loads. AFRP has been widely used in Japan in retrofitting RC structures and highway piers [187]. However, using SFRP for seismic upgrades has caused several issues, mostly with fire problems [163]. Textile reinforced mortar (TRM) is another potential option to strengthen building elements in the seismic region [188]. TRM performs better than FRP as a strengthening materials at high temperatures [189].

4.5.2. Blast load

The recent rise of blast/malicious explosions worldwide has spurred the development of new materials for refurbishment and rehabilitation of infrastructures. CFRP, GFRP, and AFRP enhancement is the most widely used methodology for improving the blast load resistance performance of structural components [190–192]. GFRP is one of the most common building materials used to improve the out-of-plan capacity of structural components such as walls and platforms for dynamic loads [191]. CFRP also improves the member's ductility, and it helps to increase blast resistance. Further, the higher tensile strength of aramid is also another option for blast retrofitting [193–195].

Liu et al. [196] studied the dynamic response and damage mechanism of CFRP protective RC piers during contact explosion and showed that the CFRP sheet could protect the RC pier. Their study also reported that wrapping columns with more layers of CFRP sheets resulted in the reduction of concrete damage due to blast load. Maazoun et al. [197] experimentally proved the efficiency of using externally bonded CFRP reinforcement on RC slabs under blast loads. Their study found that CFRP strips can be used to increase the flexural capacity and stiffness of RC. The retrofitted slab with one CFRP strip shows a higher number of tensile cracks with smaller widths. The flexural cracks are almost invisible in the slab with two CFRP strips, and no cracks are observed when using four CFRP strips [197]. It was concluded that flexural cracks on the non-loaded side of the slab could be reduced by increasing the amount of the CFRP strips. However, it will increase the damage on the blast-loaded side of the slab.

5. Applications of NFRPs in infrastructures

Architects and engineers are increasingly focusing on sustainability in contemporary practice. Natural fibers have been successfully employed to replace synthetic fibers, and new types of resin-based NFRPs have been brought to the market to lower environmental impact and embodied energy [167,198]. Using NFRP composite can be a viable option for achieving a circular economy and a sustainable environment [199]. A study by Smits et al. [167] used NFRP in the load-bearing footbridge. The bridge is built from flax fibers, bio-based resins, polylactic acid (PLA) foam, and natural cork. The fibers utilised were partially hemp and partially flax, and they were mixed with an epoxy resin having a 56% bio-content. PLA, commonly known as polylactide, is an aliphatic thermoplastic polyester manufactured from renewable resources that served as the bridge's nonstructural core [200]. This bridge's innovation will serve as a global symbol for the usage of NFRP in infrastructure [201].

The innovative bio-composite bicycle swing bridge “Ritsumasy” in the Netherlands is another remarkable structure constructed with NFRP [202]. This bridge is made of flax-reinforced epoxy. Ritsumasy Bridge demonstrates that flax composites are a good, long-term solution for CO₂ reduction and a feasible, ecologically beneficial option. This bridge was the first tangible demonstration that bio-composites may be used in building instead of wood, concrete and steel [203]. The NFRPs can also use to improve the appearance of the building using the concept of “vernacular architecture” [204]. Screens, wall and roof cladding and flooring can be made from NFRPs [205–207]. The Happyland Townhouse in Thailand uses banana fibers for external cladding to accentuate its esthetic appearance [204]. Bambach et al. [208] investigated the utilization of flax, jute, and hemp fiber composites as major structural components in civil infrastructure applications. Flax, jute, and hemp fibers were used to create natural fiber composite plates and channel sections. Their study concluded that buckling and post-buckling reactions are stable, and failure occurs in a slow and ductile manner.

Furthermore, Bambach et al. [208] research study also promote the usage of NFRP composite in light structural applications, such as residential and commercial buildings. Another study conducted by Bambach et al. [209] compared the performance of NFRPs and SFRPs in composite structural components [208]. In terms of compression strength and stiffness, the NFRP structural laminates have proved to be structurally comparable to the SFRP composites. NFRPs are considered as feasible alternatives to synthetic geotextile fibers because of their strong mechanical characteristics, low cost, and ability to break down into non-polluting components [210]. Jute and coir fibers have emerged as the ideal research materials for natural geotextiles [211]. Geotextiles are the most widely used geotechnical engineering as reinforcement, filtration, drainage, stabilization, barrier, and erosion protection [101,212].

Natural geotextiles have also been employed in road infrastructure due to their high initial tensile strength [211]. Roads made of jute geotextiles have been built in various parts of Bangladesh, and jute geotextiles have a higher load-bearing capacity [213]. The subdivision of rural roads is being enhanced by an increasing number of natural geotextiles in India. This is because natural geotextiles have a high initial tensile strength, allowing the subgrade to withstand minimal stress throughout the road's construction and operation [214]. Another possible application for jute/coir geotextile is to improve the performance of river embankment earthen structures [215,216]. It's estimated that fibers of coconut coir buried in clay soil will retain 80% of their tensile strength after a half-year [217]. In addition to soil stabilization, jute fibers are also utilised for filtration and drainage purposes. This indicates that NFRP composites can be a sustainable option in road construction [218,219], retaining walls [220] and earthquake engineering [221,222]. However, more research studies are needed to ensure their structural, fire and durability performance.

6. Applications of HFRPs in infrastructures

In recent years, several countries have made significant progress in the research and development of all HFRP structures in infrastructure [223]. The first HFRP pedestrian bridge was built (Fig. 13a) in Okinawa, Japan [224]. All the structural elements have been made with GFRP at the beginning stage because a heavily corrosive environment surrounds it. GFRP is less expensive than CFRP but has less stiffness and strength. There may be challenges in employing 100% GFRP for this bridge since the deflection ratio to span is a significant problem in bridge design. One solution to the aforementioned difficulty is to combine GFRP with the high strength and stiffness of CFRP. Therefore, an HFRP solution (combination of GFRP and CFRP) was chosen for this bridge design [225]. The benefit of utilising this hybridisation is allowing to use of high-strength CFRP in the flanges while keeping material costs low by using GFRP in the flanges and web. Further, the fatigue performance can be improved by incorporating a suitable proportion of CFRP/GFRP into a hybrid composite. Hai et al. [225] experimentally validated the success of using HFRP with the optimum use of CFRP and GFRP in the flanges. It was proven that carbon volume content in the flanges influences the failure modes and strength of HFRP composite beams.

The main supporting beams in West Mill Bridge in the United Kingdom are another example of the application of HFRP (Fig. 12b) [226,227]. The HFRP is made with CFRP and GFRP. Recent study by Asyraf et al. [228] investigated the usage of HFRP in a cross-arm structure for transmission tower application. The HFRP composed of flax and pineapple fiber shows a similar tensile strength and stiffness as GFRP [226]. This HFRP can be a suitable sustainable alternative option for the SFRP in structural applications, particularly as cross-arm beams.



Fig. 13. Application of HFRP in infrastructure: a) Pedestrian bridge in Japan; and b) West Mill bridge in UK [227].

7. Summary and discussion

Previous studies highlighted that SFRP, NFRP and HFRP bring many benefits to infrastructure construction due to their various improvements and gains related to strength, durability, cost, sustainability, etc. However, these benefits depend on the type of fibers or blend fibers used in the infrastructure construction and context.

Overall, in terms of durability, the anti-corrosive nature of SFRP, NFRP and HFRP make them used as superior alternatives to reinforced concrete that is produced with ordinary steel bars. Moreover, the improved resistance to chlorines, alkalis, and carbonization leads the above FRPs to use in aggressive environments such as seawater. However, selecting the most appropriate FRP type for the infrastructure application is more complex. As the performance of the durability indicators varies with the application types thus, they should be chosen wisely. For example, the thickness swelling is an important indicator of the durability of concrete, and it is low for NFRPs compared with SFRPs. Further, previous studies also state that the NFRP gives a more persistent mechanical performance in a moist environment than GFRP [73,78].

Moreover, NFRP delivers a high confinement strength and confinement modulus, making them more ductile and preferable to use in infrastructure engineering applications, which require high energy dissipation against lateral loads. HFRP is another alternative when the durability indicators must be compromised between SFRP and NFRP. Further, when the durability indicators (i.e. resistance to chemical attack, corrosion, water absorption, freeze-thaw and fatigue) are concerned, SFRP will give a more reliable response than NFRP and vice versa where HFRP can be considered to be used.

The sustainability aspects of using different fibers are concerned; the amount of greenhouse gas emissions are significantly reduced when NFRP is used in the concrete compared to that SFRP [143]. For instance, carbon fibers emit approximately 60 times higher CO₂ equivalents than most natural fibers [153], which makes use of natural fibers promising in the future. Notably, the embodied energy of carbon fibers and other SFRP is less than the conventional steel reinforcement. Compared to conventional steel reinforcement, FRPs give a slightly lower or higher performance in terms of strength while having less detrimental effects on the environment.

Furthermore, using FRPs in retrofitting and renovating rather than demolishing the damaged existing structures saves a significant amount of raw material, time, labor and cost [4,6]. Moreover, the application of FRPs in infrastructure construction reduces the negative impacts on the environment, which is the goal of the construction industry and governments [153]. This indicates that the application FRPs in infrastructure construction enables green building construction while using NFRP is more appropriate than SFRP. Further, the negative environmental impacts induced by using SFRP can be reduced by using HFRP, as it can be produced with a combination of SFRP and NFRP.

The building infrastructure construction and maintenance should be economical and feasible when considering the cost. Lotfi et al. [229] studied the cost of various fibers and found that the unit price of most NFRPs is significantly lower than SFRPs such as carbon and glass (Fig. 14). Fig. 14 shows that NFRPs such as sisal, jute, bamboo, coir and kenaf gives very low price compared to other fibers. It should be noted that before using the fibers for any infrastructure applications, a cost-benefit analysis should be done to identify the economic feasibility aligned with mechanical, durability and sustainability aspects.

The material's resistance and resilience against fire and elevated temperatures are vital in infrastructure engineering applications. Fig. 15 shows the comparison of tensile strength of SFRPs, NFRPs and HFRPs at elevated temperatures. Ou et al. [19,139] studied the tensile behavior of glass fiber-reinforced composites at different strain rates and elevated temperatures. The study concluded that Young's modulus and the tensile strength reduce with the increasing temperature, where a maximum decrease of the tensile strength was observed between 60 °C and 100 °C. The tensile strength was almost constant from 0 °C to 60 °C apart from the smidgen fluctuations. Cao et al. [85] researched the tensile behavior of CFRP, hybrid FRP combined of Carbon with Glass (C/GFRP) and Carbon with Basalt (C/BFRP) when subjected to temperatures between 16 °C and 200 °C. It was evident that the tensile strengths of CFRP, C/GFRP

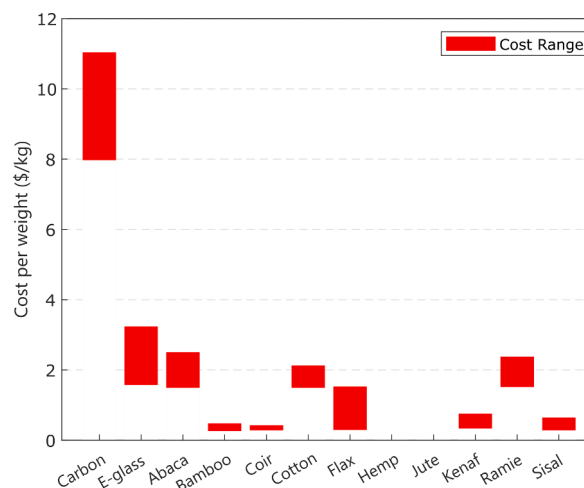


Fig. 14. Comparison of unit cost for various fiber types [229].

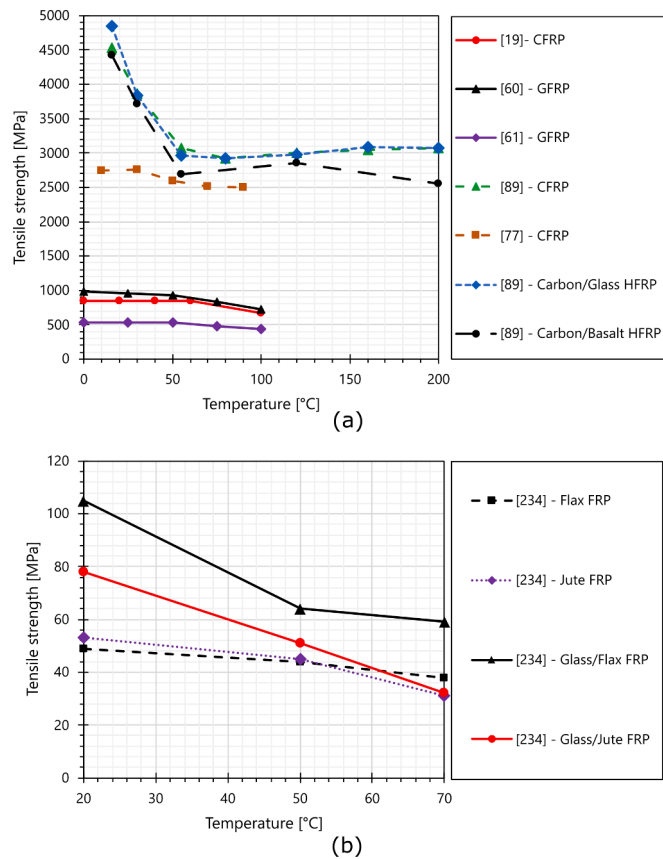


Fig. 15. Temperature effect on tensile strength of different FRPs: a) SFRP and HFRP made with SFRP; and b) NFRP and HFRP made with SFRP and NFRP.

and C/BFRP significantly reduced up to glass transition temperature (T_g) and then remained almost constant with temperature, but the tensile strength was still higher than non-impregnated FRPs at room temperature [19].

Fig. 15a shows that the tensile strengths of SFRP, NFRP, and HFRP significantly vary with increasing temperature. The tensile strength appears to decrease more (or faster) at high temperatures [83,230,231]. This could be due to the resin matrix weakening when it reached T_g [19,85,232,233]. Further, the tensile strength of NFRP was lower than that of SFRP and HFRP, which was due to the poor adhesion between natural fibers and matrix resins. However, it can be enhanced via innovative processing techniques such as chemical and physical modification procedures [107]. Cerbu et al. [234] investigated the temperature effects on the mechanical properties of hybrid FRPs with vegetable and glass fibers [234]. The tensile strength of HFRP made with GFRP and flax enable higher than the flax FRP at elevated temperatures (Fig. 15b). A similar observation was observed in the HFRP made with jute and glass FRP. Overall, the tensile strength of hybrid FRPs was higher than that of non-hybrid FRPs. Moreover, the length of the natural fiber also influences tensile strength, and long fibers have lower tensile strength than short fibers [234]. This indicates that each form of FRP has a distinct use in various infrastructures depending on the nature of the environment.

The bond performance of FRPs is an essential factor that decides their application in the infrastructure. When comparing SFRP and HFRP, the moisture absorption in NFRP resulted in poor bonding. Further, the bonding performance of SFRP is low when it reaches T_g . However, the bond strength of NFRP can be improved via modifying the fiber surface with a chemical treatment such as alkaline treatment [235]. The bonding issues related to fire on the SFRP can be modified using cement-based adhesives [236,237].

Overall, each type of fiber has benefits and limitations, but they can be employed in infrastructure construction with appropriate methods to get the most benefits for each fiber. Further, FRPs offer a high potential for continuous integration into strengthening RC structures. These materials offer exceptional characteristics and may consequently be used in a wide range of infrastructure construction applications. Future research on NFRP and HFRP composites should concentrate on environmental evaluation, durability, mechanical characteristics improvement, and moisture resistance.

8. Conclusions

Restoring existing and deteriorating infrastructures and the ability to improve the strength and durability of structures with lower weight and easy installation are the major advantage of FRP. A comprehensive review of tensile, fire, bond, durable and sustainable performance of SFRP, NFRP and HFRP and the current state in trending applications in infrastructures was carried out. The challenges

and the benefits that are attributed to the application of SFRP, NFRP and HFRP in infrastructure were also discussed. Based on this study, the following conclusions are reached:

- The tensile strength of concrete can be increased when FRP is added to the concrete, and the level of strength increment depends on FRP types (i.e. SFRP, NFRP and HFRP), fiber properties, and cement matrix and dosage level. This increased tensile characteristic of FRP brings insights into reducing the amount of steel reinforcement in concrete and improving crack resistance.
- The strength reduction at elevated temperatures has hindered the usage of FRP in infrastructure applications. CFRP and GFRP lose about 50% of their strength at elevated temperatures of 250 °C and 325 °C, respectively. Furthermore, the strength and stiffness of both GFRP and CFRP significantly reduced when they were exposed to over 400 °C. It was amply proved that the usage of cement-based adhesive with various materials could be a sustainable method to avoid resin exceeding T_g. Even though NFRP bring in a lot of advantages, one significant disadvantage of NFRP is their low fire resistance. The decomposition process happens at temperatures ranging from 300 °C to 500 °C, producing flammable gas, char, and smoke.
- The durability of concrete under harsh environmental conditions and service loads can be improved when it contains FRPs compared to conventional reinforced concrete. Using NFRP and SRP in concrete reduces the durability effects due to corrosion. However, resistance to alkaline and acid attack is still problematic, even though FRP increases the resistance to alkaline and acids to certain levels.
- Using FRP in the form of synthetic and natural fiber in the building sector has been shown to reduce environmental impacts. Therefore, FRP materials are considered to be a vital component for the sustainable growth of the building sector while being a low-cost alternative to conventional reinforcement and should be appropriately marketed as an environmental solution in the present global context.
- HFRP composites are an emerging area in civil engineering that is growing rapidly for use in a variety of infrastructure applications. HFRP is a promising alternative to SFRP and NFRP to improve the strength, fire, durable and sustainable infrastructure performance.

CRedit authorship contribution statement

Satheeskumar Navaratnam: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Supervision. **Kajanan Selvaranjan:** Methodology, Formal analysis, Investigation, Writing – original draft. **Darshana Jayasooriya:** Formal analysis, Investigation, Writing – original draft. **Pathmanathan Rajeev:** Conceptualization, Resources, Data curation, Writing – review & editing, Supervision. **Jay Sanjayan:** Resources, Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A

Table A1
Flexural strength performance of SFRP.

Researchers	Strengthening system	Application	Results
Haddad et al. [238]	NSM CFRP strip	Thermally damaged concrete beams	The use of CFRP in thermally damaged beams improves load capacity while reducing ductility. End-cover separation was the failure mode.
Zhang et al. [239]	NSM CFRP and GFRP bars	RC T- section beams	A general increase in flexural stiffness of the strengthened specimens was observed, especially at the post yield stage of loading. The flexural stiffness, yield load, and ultimate load of CFRP-reinforced specimens were consistently greater than those of GFRP-reinforced specimens. End anchorage was critical in preventing the NSM FRP bars from debonding prematurely. It can be seen that as the diameter of the bar increased, the specimen's ultimate load went up as well. The stiffness of the specimens, on the other hand, was practically consistent. The strain demand on the steel bars was reduced by increasing the FRP-NSM bar diameter.
Yeboah et al. [46]	NSM BFRP and GFRP bars	Timber beams	Brittle tensile failure of the timber in the tensile zone was the main failure mechanism identified for both NSM FRP reinforced and unreinforced specimens. The NSM reinforcement resulted in a significant increase in the ultimate load (33–69%) and flexural stiffness (22–33%) of the timber beams.

(continued on next page)

Table A1 (continued)

Researchers	Strengthening system	Application	Results
Panahi et al. [7]	EB SFRP sheets and NSM SFRP bars	RC beams	In comparison to the control beam, the flexural capacity and stiffness of reinforced concrete beams with NSM FRP rods increased significantly, while the mid-span deflections of strengthened beams at failure mainly decreased. All reinforced specimens with FRP composites displayed a similar load-deflection behavior, with an approximate tri-linear response described by concrete cracking, steel yielding, and post-yielding stages up to failure mode. The ultimate bending moment of reinforced beams was approximately 3.28–5.56 times that of the control beam.
Stylianidis et al. [90]	SFRP plate/laminate	Steel-concrete composite beams	The ultimate curvature of the critical cross-section, as well as the corresponding bending moment capacity of the FRP-strengthened steel-concrete section, have been proposed. The tensile capacity of the FRP component can only expand up to a certain limit, which is determined by the ratio of the steel beam section's tensile capacity to the concrete slab's compressive resistance.

Table A2

Different types of adhesives to avoid the effect of fire.

References	Type of adhesive	Raw materials	Performances of adhesives
Alwash et al. [236]	Nano modified cement-based adhesive	Graphene oxide, graphene nanoplatelets and nano silica	1. Compression and flexural strength improved by up to 46% and 57%, respectively 2. 61% lower chloride penetration 3. 27% and 73% improvement in bond properties for CFRP laminate and textile, respectively
Ding et al. [240]	Phosphate cement-based adhesive	Potassium dihydrogen phosphate, magnesia, and fly ash	1. About 47% improved fire resistance 2. Enable high flexural strength 3. Significant compressive strength
Menna et al. [241]	Metakaolin based geopolymer adhesive	Metakaolin powder, NaOH pellets, sodium silicate solution, pure quartz powder	1. Good adhesion to concrete substrate and to steel cords 2. Poor adhesion between carbon and geopolymer 3. Increases in the ultimate bending moment
Nguyen et al. [242]	Nanoparticle based epoxy adhesive	Nano-SiO ₂ , nano-Fe ₂ O ₃ , nano- clay and nano-TiO ₂	1. Improved adhesion strength 2. Improved glass transition temperature up to 170 °C 3. Enhanced the impact strength
Lei et al. [243]	Montmorillonite nano clay modified siloxane epoxy adhesive	Montmorillonite nano clay	1. Improved the mechanical performance of FRP-concrete interface 2. Slowed down the ingress of water and chloride ions in the aggressive environment 3. Improved the thermal stability
Firouz et al. [237]	Surface Mount epoxy Adhesive	Sulphate resistant cement and 2 mm aggregates	1. Excellent fire rating 2. Good bond performance 3. Adequate tensile capacity
Zhang et al. [244]	Magnesium phosphate cement epoxy Adhesive	Silica fume, calcium carbonate whisker, magnesium phosphate cement	1. Improves bond capacity 2. Improves bond ductility
Yang et al. [245]	Engineered cementitious composites (ECC)	Cement, fly ash, silica fume and polyvinyl alcohol (PVA) fiber	1. Improved load bearing capacity with CFRP and BFRP rods 2. Ductile failure behavior
Kuntal et al. [246]	High strength cement grout (HSCG) adhesive	Quick-set non-shrink grout used	1. Adequate bond strength 2. Improved the peak strength with beams without stirrups
	Geopolymer mortar adhesive	Ground granulated blast furnace slag, sand, sodium hydroxide, sodium silicate solution	1. Adequate bond strength 2. Improved the peak shear strength and ultimate displacement of beams without stirrups
Yang et al. [247]	ECC adhesive	Cement, fly ash, silica fume and polyvinyl alcohol (PVA) fiber	1. More ductile failure due to the multiple cracking behavior in ECC 2. Usage of ECC with epoxy good at debonding control
Toutanji et al. [248]	Inorganic epoxy adhesive	Aluminosilicate powder, water-based activator	1. Adequate ultimate strain of CFRP reinforcement with epoxy

Table A3

Strengthening technique of natural fibers in timber elements.

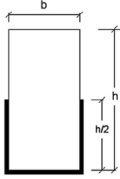
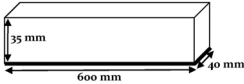
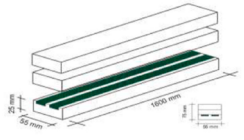
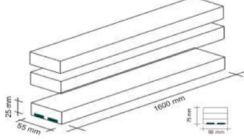
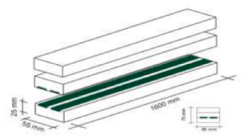
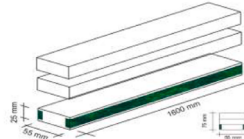
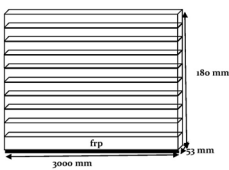
References	Type of FRP	Strengthening technique	Test method	Failure mode	Tensile/ Flexural strength	Bearing capacity
García et al. [249]	Basalt fibers		Single point of load testing	Failure of the lower fibers when in tension, or due to a combination of tension and shear	21% ↑	24.5% ↑
Wang et al. [114]	Flax fibers		Three-point bending	Tensile failure and debonding at mid-span	↑	60.7% ↑
Khadaf et al. [115]	Bamboo strips		Two point of load testing	-	25% ↑	24.8% ↑
				-	4.3% ↑	4.1% ↑
				-	6.4% ↑	6.3% ↑
				-	1.93% ↑	1.7% ↑
Mascia et al. [113]	Sisal fibers		Single point of load testing	Failure of the tensile lamella	30.07% ↑	22% ↑

Table A4

SFRPs application in bridge structures

References	Country	Period	Elements of bridges used SFRP	Strengthening system	Application
[167]	Netherlands	1995–2016	Main load-bearing structure, Modular edge elements, Bridge deck systems	Unidirectional CFRP, Unidirectional GFRP	SFRP footbridge in Harlingen, Fly-over Waarderpolder in Haarlem, Juliana Bridge in Zaanstad, Wildlife crossing and tunnel in Rijnssse-Wierden, Highway N201 viaducts at Schiphol
[38,169, 170]	USA	1990–2016	Bridge girders, Bridge decks and piers, Concrete pavement, Floating bridge, Fender piles, Bridge drains, Shear walls, Bent cap wrapping, Culvert invert liners	GFRP bars, CFRP and GFRP sheets, GFRP wrapping,	SFRP deck at No-Name Creek, SFRP sandwiched deck, Schuyler Heim Bridge, Chemung County Bridge, Sauvie Island Bridge, Allen Creek, Gandy, and Friendship Trails Bridges, Powder Point Bridge, Edges of four bridges in St. Louis, Martin Spring Bridge, Wilson's Bridge, Schroon River Truss Bridge, Kings Stormwater Canyon Bridge, Bennet's Bridge

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Table A4 (continued)

References	Country	Period	Elements of bridges used SFRP	Strengthening system	Application
[39,169, 187]	Japan	1996–2015	Bridge girders and bridge decks	Pultruded GFRP	Okinawa Road Park Bridge, Tskuba Foot Bridge
[170,250, 251]	UK	1991–2017	Cables and deck, Complete bridge, Deck, Deck and beams, Deck and girder, pier and trestles, Parapet replacement, Superstructure replacement, Bridge deck slabs,	GFRP fiber and Kevlar-49 fiber	Aberfeldy Footbridge, Bonds Mill Lift Bridge, Parson's Footbridge, Halgavor Bridge, West Mill Bridge, Mount Pleasant Bridge, St Austell Railway Bridge, Launder Aqueduct, Wilcott Bridge, Mort Lane Parapet, River Leri Footbridge, Staden Hay, Bradkirk Footbridge, Thompson's Bridge Deck Slabs, Calder & Rubha Gas Viaducts, Moss Canal Bridge, Dawlish Station Footbridge, Dragon Bridge, Purfleet Footbridge, FRP Parapets, Church Road Bridge, River Chor Aqueduct, Thornaby Footbridge, Sedlescombe Bridge, Bull Ring Farm Road Overbridge, Mapledurham Bridge, Bird Riding Footbridge, East Row Footbridge, Emersons Green East Cycle, Dover Sea Wall, Kiora Sluice Footbridge, Prince Street Footbridge
[169,252] [253]	China New Zealand	– 2004–2020	Deck and girders Shear walls	GFRP fiber High strength carbon fiber strips, Carbon fiber plates, Unidirectional CFRP strips, GFRP bars, CFRP fabric, CFRP laminate	FRP cable-stayed bridge in Jiangsu University, Miyun bridge, Grafton Bridge, Port Hills Overbridge, MacKays to Peka Peka express bridge,
[254]	Australia	1990–2017	Concrete beams, headstocks, deck,		Melbourne's Westgate Bridge, pedestrian bridge in Centennial Pioneer Park in Perth, Brisbane City Council - Bowen Bridge, Victoria Little River Bridge & Warrigal, Overpass, Hobsons Bay & Melbourne City Councils, Kororoit Creek Rd Bridge & Arden St Bridge, Melbourne Water – Maribymong River Pipe Bridge, Albury/Wodonga Rail Bridge, William Slim Bridges, Ouse River Bridge, Emu River Bridge, Greenough River Bridge

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